

How to use these token cutouts

These pages contain the text *The Cat in the Hat* by Dr Seuss (1832 tokens, entropy 0.8 bits/token, perplexity 1.7). Each **cutout** is a **token** preceded by its **prefix**—the 2 tokens that came immediately before it in the original text.

Anatomy of a cutout

the sun

prefix (2 tokens)

did

token (the next word that followed)

Every distinct token has its own colour. **Prefix** words appear inside a coloured box (the prefix word’s own colour as the background, with white text); the free-standing **token** appears in plain coloured text. The same word always wears the same colour, whether you see it inside a box or as a token. Two unrelated tokens can occasionally share a colour, so always verify the word itself matches—not just the colour.

Setup

Cut out the tokens along the dotted lines and **spread them out** face-up on the table, with no overlap if possible. This is the “training” step: every prefix-token combination from the original text is now a physical cutout sitting on your table.

How to play: the matching game

To generate text, repeatedly find a cutout whose **prefix** matches the last 2 tokens you’ve written, then **write its token** onto your page. The token you just wrote becomes part of the prefix you’ll match against next—that’s the chain.

Worked example

Step 1. Pick any cutout to begin—say this one:

the sun did

Copy its prefix *and* its token onto your page. Your text so far:

the sun did

Step 2. Look at the last 2 tokens you’ve written (in this case *sun did*) and find a cutout whose prefix matches them, like this one:

sun did not

Write its token onto your page. Your text now reads:

the sun did not

Step 3. Repeat. The last 2 tokens you’ve now written are *did not*—find another matching cutout:

did not shine

Add its token. Your text now reads:

the sun did not shine

Keep going—write as much or as little as you like.

Tips

- **Use colour as a fast filter.** Scan by the rightmost prefix box’s colour first, then verify the actual word matches.
- **Pick from many candidates by eye.** The more cutouts share a given prefix, the more often your eye lands on one—that’s **weighted sampling** for free, and it’s why some words follow others more often in your text.
- **Partial matches are OK.** If no cutout matches all your last 2 tokens, settle for one where just the rightmost prefix token matches.
- **Put the cutout back** after using it—removing it would change the model’s distribution next time round.

the sun the sun did sun did not did not shine
not shine . shine . it . it was it was too
was too wet too wet to wet to play to play .
play . so . so we so we sat we sat in sat in the
in the house the house all house all that
all that cold that cold , cold , cold ,
cold , wet , wet day wet day . day . I . I sat
I sat there sat there with there with Sally
with Sally . Sally . we . we sat we sat there
sat there , there , we , we two we two .
two . and . and I and I said I said , said , how

, how I how I wish I wish we wish we had
we had something had something to
something to do to do too do too wet
too wet to wet to go to go out go out and
out and too and too cold too cold to
cold to play to play ball play ball . ball . so
 . so we so we sat we sat in sat in the
in the house the house . house . we . we did
we did nothing did nothing at nothing at all
at all . all . so . so all so all we all we could
we could do could do was do was to was to sit

to sit sit sit sit sit sit sit sit and
sit and we and we did we did not did not like
not like it like it . it . not . not one
not one little one little bit little bit . bit . and
. and then and then something
then something went something went bump
went bump how bump how that
how that bump that bump made
bump made us made us jump us jump we
jump we looked we looked then
looked then we then we saw we saw him

saw him step him step in step in on in on the
on the mat the mat we mat we looked
we looked and looked and we and we saw
we saw him saw him the him the cat
the cat in cat in the in the hat the hat and
hat and he and he said he said to said to us
to us , us , Why , Why do Why do you
do you sit you sit there sit there like
there like that like that I that I know
I know it know it is it is wet is wet and
wet and the and the sun the sun is sun is not

is not sunny not sunny . sunny . But . But we
But we can we can have can have Lots
have Lots of Lots of good of good fun
good fun that fun that is that is funny
is funny I funny I know I know some
know some good some good games
good games we games we could we could play
could play , play , said , said the said the cat
the cat . cat . I . I know I know some
know some new some new tricks new tricks ,
tricks , said , said the said the cat the cat in

cat in the in the hat the hat . hat . a . a lot
a lot of lot of good of good tricks good tricks .
tricks . I . I will I will show will show them
show them to them to you to you . you . your
 . your mother your mother will
mother will not will not mind not mind at
mind at all at all if all if I if I do I do .
do . then . then Sally then Sally and
Sally and I and I did I did not did not know
not know what know what to what to say
to say . say . our . our mother our mother was

mother was out was out of out of the
of the house the house For house For the
For the day the day . day . But . But our
But our fish our fish said fish said , said , no
, no no no Make no Make that
Make that cat that cat go cat go away
go away tell away tell that tell that cat
that cat in cat in the in the hat the hat you
hat you do you do not do not want
not want to want to play to play . play . he
. he should he should not should not be

not be here be here . here . he . he should
he should not should not be not be about
be about . about . he . he should he should not
should not be not be here be here When
here When your When your mother
your mother is mother is out is out now
out now now now have now have no
have no fear no fear . fear . have . have no
have no fear no fear said fear said the
said the cat the cat . cat . my . my tricks
my tricks are tricks are not are not bad

not bad , bad , said , said the said the cat
the cat in cat in the in the hat the hat .
hat . Why . Why , Why , we , we can
we can have can have Lots have Lots of
Lots of good of good fun good fun , fun , if
, if you if you wish you wish , wish , with
, with a with a game a game that game that I
that I call I call up call up up up up up
up up with up with a with a fish a fish put
fish put me put me down me down said
down said the said the fish the fish . fish . this

.this is this is no is no fun no fun at
fun at all at all put all put me put me down
me down said down said the said the fish
the fish . fish . I . I do I do not do not wish
not wish to wish to fall to fall have
fall have no have no fear no fear said
fear said the said the cat the cat . cat . I
. I will I will not will not let not let you
let you fall you fall . fall . I . I will I will hold
will hold you hold you up you up high
up high as high as I as I stand I stand on

stand on a on a ball a ball . ball . with
 . with a with a book a book one book one one
one one hand one hand and hand and a
and a cup a cup on cup on my on my hat
my hat But hat But that But that is that is not
is not all not all I all I can I can do
can do said do said the said the cat the cat .
cat look . look at look at me
at me look me look at look at me at me now
me now said now said the said the cat
the cat . cat . with . with a with a cup

a cup and cup and a and a cake a cake on
cake on the on the top the top of top of my
of my hat my hat I hat I can I can hold
can hold up hold up two up two books
two books I books I can I can hold
can hold up hold up the up the fish
the fish and fish and a and a little a little toy
little toy ship toy ship and ship and some
and some milk some milk on milk on a
on a dish a dish and dish and look and look I
look I can I can hop can hop up hop up and

up and down and down on down on the
on the ball the ball But ball But that
But that is that is not is not all not all Oh
all Oh , Oh , no , no . no . that . that is
that is not is not all not all . all look
. look at look at me at me look me look at
look at me at me look me look at look at me
at me now me now it now it is it is fun
is fun to fun to have to have fun have fun But
fun But you But you have you have to
have to know to know how know how .

how . I . I can I can hold can hold up

hold up the up the cup the cup and

cup and the and the milk the milk and

milk and the and the cake the cake I

cake I can I can hold can hold up

hold up these up these books these books and

books and the and the fish the fish on

fish on a on a rake a rake I rake I can

I can hold can hold the hold the toy

the toy ship toy ship and ship and a

and a little a little toy little toy man

toy man and man and look and look with
look with my with my tail my tail I tail I can
I can hold can hold a hold a red a red fan
red fan I fan I can I can fan can fan with
fan with the with the fan the fan as fan as I
as I hop I hop on hop on the on the ball
the ball But ball But that But that is
that is not is not all not all . all . Oh . Oh ,
Oh , no , no . no . that . that is that is not
is not all not all . all that . that is
that is what is what the what the cat

the cat said cat said . said then
 . then he then he fell he fell on fell on his
 on his head his head he head he came
 he came down came down with down with a
 with a bump a bump From bump From up
 From up there up there on there on the
 on the ball the ball . ball . and . and Sally
 and Sally and Sally and I and I , I , we
 , we saw we saw all saw all the all the things
 the things fall things fall and fall and our
 and our fish our fish came fish came down

came down , down , too , too . too . he
 . he fell he fell into fell into a into a pot
 a pot he pot he said he said , said , do , do I
 do I like I like this like this Oh this Oh ,
 Oh , no , no I no I do I do not do not .
 not . this . this is this is not is not a
 not a good a good game good game ,
 game , said , said our said our fish our fish as
 fish as he as he lit he lit . lit . no . no , no , I
 , I do I do not do not like not like it like it ,
 it , not , not one not one little one little bit

little bit now bit now look now look what
look what you what you did you did said
did said the said the fish the fish to fish to the
to the cat the cat . cat . now . now look
now look at look at this at this house
this house look house look at look at this
at this look this look at look at that
at that you that you sank you sank our
sank our toy our toy ship toy ship ,
ship , sank , sank it sank it deep it deep in
deep in the in the cake the cake . cake . you

.you shook you shook up shook up our
up our house our house and house and you
and you bent you bent our bent our new
our new rake new rake . rake . you

.you should you should not should not be
not be here be here When here When our
When our mother our mother is mother is not
is not . not . you . you get you get out
get out of out of this of this house
this house said house said the said the fish
the fish in fish in the in the pot the pot .

pot. But . But I But I like I like to like to be
to be here be here . here . Oh . Oh , Oh , I
, I like I like it like it a it a lot a lot said
lot said the said the cat the cat in cat in the
in the hat the hat to hat to the to the fish
the fish in fish in the in the pot the pot .

pot. I . I will I will not will not go
not go away go away . away . I . I do I do not
do not wish not wish to wish to go to go and
go and so and so , so , said , said the
said the cat the cat in cat in the in the hat

the hat , hat , so , so so so so so so . so .
.. . . I . I will I will show will show you
show you another you another good
another good game good game that
game that I that I know I know and
know and then and then he then he ran
he ran out ran out . out . and . and ,
and , then , then , then , fast , fast as fast as a
as a fox a fox , fox , the , the cat the cat in
cat in the in the hat the hat came
hat came back came back in back in with

in with a with a box a box . box . a . a big
a big red big red wood red wood box
wood box . box . it . it was it was shut
was shut with shut with a with a hook
a hook . hook . now . now look now look at
look at this at this trick this trick , trick , said
, said the said the cat the cat . cat . take
. take a take a look a look then look then he
then he got he got up got up on up on top
on top with top with a with a tip a tip of
tip of his of his hat his hat . hat . I . I call

I call this call this game this game fun
game fun in fun in a in a box a box ,
box , said , said the said the cat the cat .
cat . in . in this in this box this box are
box are two are two things two things I
things I will I will show will show to
show to you to you now you now . now . you
 . you will you will like will like these
like these two these two things two things ,
things , said , said the said the cat the cat with
cat with a with a bow a bow . bow . I . I will

I will pick will pick up pick up the
up the hook the hook . hook . you . you will
you will see will see something
see something new something new . new . two
. two things two things . things . and . and I
and I call I call them call them Thing
them Thing one Thing one and one and Thing
and Thing two Thing two . two . these
. these things these things will things will not
will not bite not bite you bite you . you . they
. they want they want to want to have

to have fun have fun . fun . then . then ,
then , out , out of out of the of the box
the box came box came Thing came Thing two
Thing two and two and Thing and Thing one
Thing one and one and they and they ran
they ran to ran to us to us fast us fast .
fast . they . they said they said , said , how
, how do how do you do you do you do would
do would you would you like you like to
like to shake to shake hands shake hands with
hands with Thing with Thing one

Thing one and one and Thing and Thing two
Thing two and two and Sally and Sally and
Sally and I and I did I did not did not know
not know what know what to what to do
to do . do . so . so we so we had we had to
had to shake to shake hands shake hands with
hands with Thing with Thing one
Thing one and one and Thing and Thing two
Thing two . two . we . we shook
we shook their shook their two
their two hands two hands . hands . But

. But our But our fish our fish said fish said ,
said , no , no no no those no those things
those things should things should not
should not be not be in be in this
in this house this house Make
house Make them Make them go them go they
go they should they should not should not be
not be here be here When here When your
When your mother your mother is
mother is not is not put not put them
put them out them out put out put them

put them out them out said out said the
said the fish the fish in fish in the in the pot
the pot . pot . have . have no have no fear
no fear , fear , little , little fish little fish ,
fish , said , said the said the cat the cat in
cat in the in the hat the hat . hat . these
. these things these things are things are good
are good things good things . things . and
. and he and he gave he gave them
gave them a them a pat a pat . pat . they
. they are they are tame are tame . tame . Oh

. Oh , Oh , so , so tame so tame they
tame they have they have come
have come here come here to here to play
to play . play . they . they will they will give
will give you give you some you some fun
some fun on fun on this on this wet this wet ,
wet , wet , wet , wet , wet day wet day .
day . now . now , now , here , here is
here is a is a game a game that game that they
that they like they like , like , said , said the
said the cat the cat . cat . they . they like

they like to like to fly to fly kites fly kites ,
kites , said , said the said the cat the cat in
cat in the in the hat the hat . hat . no . no not
no not in not in the in the house
the house said house said the said the fish
the fish in fish in the in the pot the pot .
pot . they . they should they should not
should not fly not fly kites fly kites in
kites in a in a house a house they
house they should they should not
should not . not . Oh . Oh , Oh , the

, the things the things they things they will
they will bump will bump Oh bump Oh ,
Oh , the , the things the things they
things they will they will hit will hit Oh
hit Oh , Oh , I , I do I do not do not like
not like it like it not it not one not one little
one little bit little bit then bit then Sally
then Sally and Sally and I and I saw
I saw them saw them run them run down
run down the down the hall the hall .
hall . we . we saw we saw those saw those two

those two things two things bump
things bump their bump their kites
their kites on kites on the on the wall
the wall bump wall bump Thump
bump Thump Thump Thump Thump bump
Thump bump down bump down the
down the wall the wall in wall in the
in the hall the hall . hall . Thing . Thing two
Thing two and two and Thing and Thing one
Thing one they one they ran they ran up
ran up they up they ran they ran down

ran down on down on the on the string
the string of string of one of one kite
one kite we kite we saw we saw mother's
saw mother's new mother's new gown
new gown her gown her gown her gown with
gown with the with the dots the dots that
dots that are that are pink are pink ,
pink , white , white and white and red
and red . red . then . then we then we saw
we saw one saw one kite one kite bump
kite bump on bump on the on the head

the head of head of her of her bed
her bed then bed then those then those things
those things ran things ran about
ran about with about with big with big bumps
big bumps , bumps , jumps , jumps and
jumps and kicks and kicks and with
and with hops with hops and hops and big
and big thumps big thumps and thumps and all
and all kinds all kinds of kinds of bad
of bad tricks bad tricks . tricks . and . and I
and I said I said , said , I , I do I do not

do not like not like the like the way
the way that way that they that they play
they play if play if mother if mother could
mother could see could see this see this ,
this , Oh , Oh , Oh , what , what would
what would she would she say she say then
say then our then our fish our fish said
fish said , said , look , look look look look and
look and our and our fish our fish shook
fish shook with shook with fear with fear .
fear . your . your mother your mother is

mother is on is on her on her way

her way home way home do home do you

do you hear you hear Oh hear Oh , Oh , what

, what will what will she will she do she do to

do to us to us what us what will what will she

will she say she say Oh say Oh , Oh , she

, she will she will not will not like not like it

like it to it to find to find us find us this

us this way this way so way so do

so do something do something fast

something fast said fast said the said the fish

the fish . fish . do . do you do you hear
you hear I hear I saw I saw her saw her .
her . your . your mother your mother your
mother your mother your mother is
mother is near is near so near so , so , as
, as fast as fast as fast as you as you can
you can , can , Think , Think of
Think of something of something to
something to do to do you do you will
you will have will have to have to get
to get rid get rid of rid of Thing of Thing one

Thing one and one and Thing and Thing two
Thing two so two so , so , as , as fast
as fast as fast as I as I could I could , could , I
, I went I went after went after my
after my net my net . net . and . and I
and I said I said , said , with , with my
with my net my net I net I can I can get
can get them get them I them I bet I bet .
bet . I . I bet I bet , bet , with , with my
with my net my net , net , I , I can I can get
can get those get those things those things yet

things yet then yet then I then I let I let down
let down my down my net my net . net . it
. it came it came down came down with
down with a with a PLOP a PLOP and
PLOP and I and I had I had them had them at
them at last at last those last those two
those two things two things had things had to
had to stop to stop . stop . then . then I
then I said I said to said to the to the cat
the cat , cat , now , now you now you do
you do as do as I as I say I say . say . you

.you pack you pack up pack up those
up those things those things and
things and you and you take you take them
take them away them away Oh away Oh dear
Oh dear said dear said the said the cat
the cat . cat . you . you did you did not
did not like not like our like our game
our game . game Oh . Oh dear
Oh dear . dear . what . what a what a shame
a shame what shame what a what a shame
a shame what shame what a what a shame

a shame then shame then he then he shut
he shut up shut up the up the things
the things in things in the in the box
the box with box with the with the hook
the hook . hook . and . and the and the cat
the cat went cat went away went away with
away with a with a sad a sad kind sad kind of
kind of look of look . look . that . that is
that is good is good , good , said , said the
said the fish the fish . fish . he . he has
he has gone has gone away gone away .

away. Yes. Yes. Yes. But. But your
But your mother your mother will
mother will come will come. come. she
. she will she will find will find this
find this big this big mess big mess and
mess and this and this mess this mess is
mess is so is so big so big and big and so
and so deep so deep and deep and so
and so tall so tall, tall, we, we can
we can not can not pick not pick it pick it up
it up. up. there. there is there is no

is no way no way at way at all at all and
all and then and then Who then Who was
Who was back was back in back in the
in the house the house Why house Why ,
Why , the , the cat the cat have cat have no
have no fear no fear of fear of this
of this mess this mess , mess , said , said the
said the cat the cat in cat in the in the hat
the hat . hat . I . I always I always pick
always pick up pick up all up all my
all my playthings my playthings and

playthings and so and so . so I

. I will I will show will show you

show you another you another good

another good trick good trick that trick that I

that I know I know then know then we

then we saw we saw him saw him pick

him pick up pick up all up all the

all the things the things that things that were

that were down were down . down . he

. he picked he picked up picked up the

up the cake the cake , cake , and , and the

and the rake the rake , rake , and , and the
and the gown the gown , gown , and , and the
and the milk the milk , milk , and , and the
and the strings the strings , strings , and
, and the and the books the books ,
books , and , and the and the dish the dish ,
dish , and , and the and the fan the fan ,
fan , and , and the and the cup the cup ,
cup , and , and the and the ship the ship ,
ship , and , and the and the fish the fish .
fish . and . and he and he put he put them

put them away them away . away . then
. then he then he said he said , said , that
, that is that is that is that . that . and
. and then and then he then he was
he was gone was gone with gone with a
with a tip a tip of tip of his of his hat
his hat . hat . then . then our then our mother
our mother came mother came in came in and
in and she and she said she said to said to us
to us two us two , two , did , did you
did you have you have any have any fun

any fun tell fun tell me tell me . me . what
 . what did what did you did you do
you do and do and Sally and Sally and
Sally and I and I did I did not did not know
not know what know what to what to say
to say . say . should . should we should we tell
we tell her tell her the her the things
the things that things that went that went on
went on there on there that there that day
that day should day should we should we tell
we tell her tell her about her about it

about it now it now , now , what , what should
what should we should we do we do Well
do Well . Well what . what would
what would you would you do you do if
do if your if your mother your mother asked
mother asked you